IIT Roorkee

DFT-D3 for Solid-State Materials

A Data-Driven Perspective on Accuracy

Presented by Tushar

Why Do We Need Better Accurate Interaction Calculations?

Fundamental Importance

Interatomic interactions determine how atoms bond and arrange in materials, directly influencing their structure and stability. Accurate modeling of these interactions is essential for predicting material properties and behavior reliably.

Role of Weak Forces

Weak interactions such as van der Waals forces play a crucial role in systems like molecular crystals, layered materials, and surface adsorption. Even though they are small in magnitude, they significantly affect stability and structural organization.

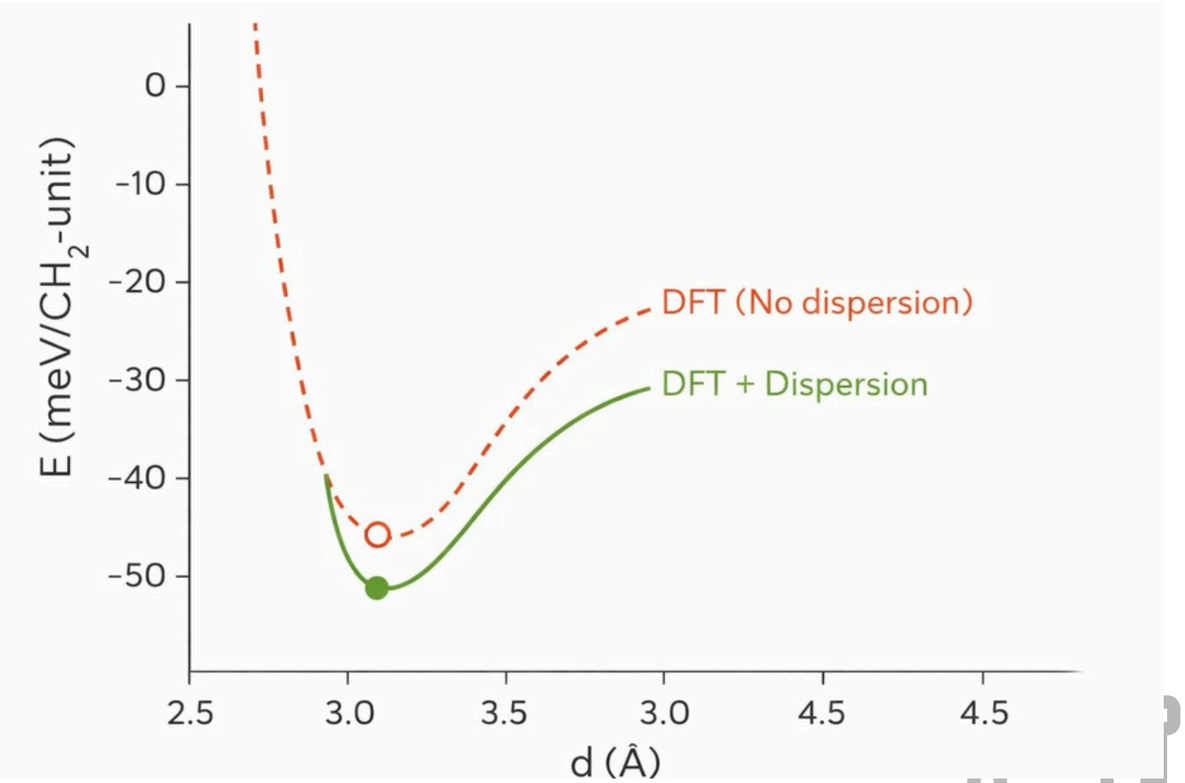
Motivation

These weak interactions are difficult to capture using conventional computational methods, which often neglect dispersion effects. This limitation motivates the development of improved approaches like dispersion-corrected DFT.

“The Missing Piece in DFT”

Dispersion interactions arise from instantaneous dipole induced dipole correlations between atoms. These interactions are inherently quantum mechanical and originate from electron correlation effects.

The strength of dispersion forces decays with interatomic distance following an inverse sixth-power law (R^{-6}), making them long-range but weak interactions that become significant in extended systems.



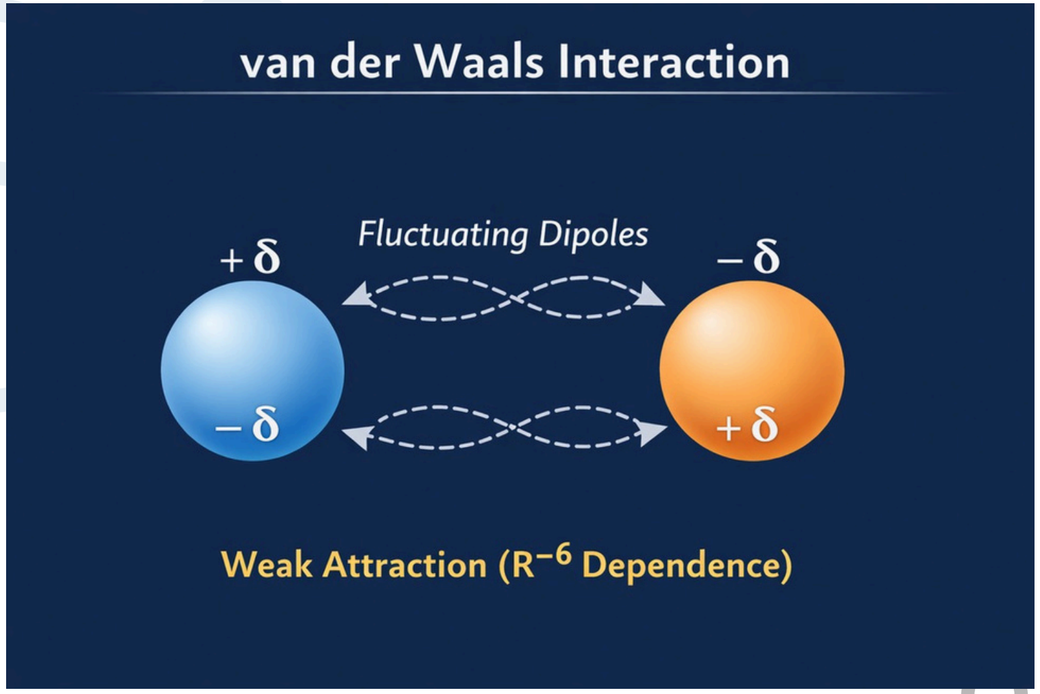
Invisible Forces: van der Waals Interaction

- **Definition :** Weak interactions between atoms/molecules
  Caused by fluctuating dipoles

- **Mechanism :** Temporary charge separation → (δ^+, δ^-)
  Induces dipole in nearby atom
  Leads to attractive force

$$E \propto 1 / R^6$$

Influences crystal stability



Adding Missing Piece : Dispersion Corrections

Problem : Standard DFT misses dispersion forces Leads to inaccurate energy predictions

$$E_{\text{Total}} = E_{\text{Base}} + E_{\text{disp}}$$

DFT energy is corrected by adding dispersion contribution

Solution : Add an extra dispersion correction term Improves accuracy of calculations

Common Method : DFT-D3 correction widely used Accounts for long-range interactions

Evolution of Grimme's DFT-D Methods

Background Dispersion corrections developed to improve DFT accuracy Proposed by Grimme

Evolution of Methods : D2 → Basic pairwise correction D3 → Improved accuracy with environment dependence D3(BJ) → Better short-range behavior (Becke–Johnson damping)

Why Improvement Needed? To achieve higher accuracy To handle different material environments

Key Advantage Computationally efficient and accurate. And Widely used in solid-state calculations

Progressive improvement in dispersion correction methods

DFT-D2 Model (Grimme)

- **Basic Idea :** First dispersion correction method Adds simple pairwise interaction
- **Key Feature :** Uses fixed parameters for all atoms No environment dependence
- **Limitation :** Less accurate for complex systems Cannot adapt to different bonding environments

The expression for D2-dispersion correction is given as:

$$E_{\text{disp}}^{\text{D2}} = -s_6 \sum_{A<B} \frac{C_6^{AB}}{R_{AB}^6} f_{\text{damp}}(R_{AB})$$

with

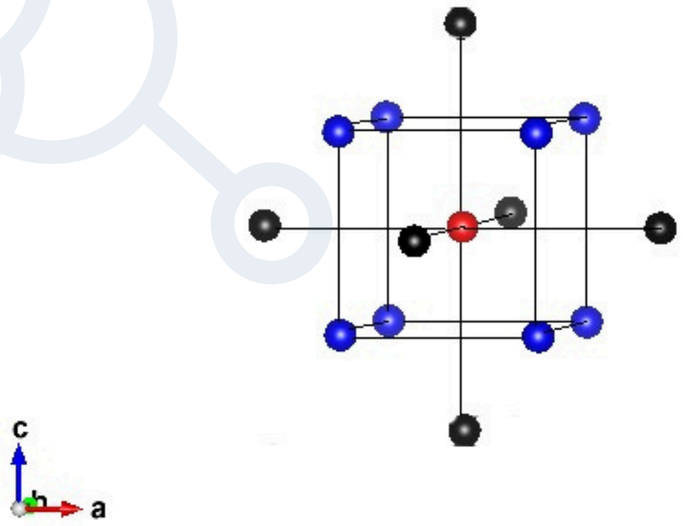
$$C_6^{AB} = \sqrt{C_6^A C_6^B}, \qquad C_6^a = 0.05\, N_p^a\, I_p^a\, \alpha^a$$

Npa represents the effective number of electrons,
Ipa donates the ionization potential,
αa refers to the static polarizability of atom aaa

DFT-D3 Model

Better accuracy due to environment awareness

- **Improvement over D2** Includes environment-dependent parameters
- **Key Feature** Uses coordination number (CN) Adapts to atomic surroundings
- **Advantage** More accurate for solid-state materials



The expression for D3 has two body terms $E^{(2)}$ and three body terms $E^{(3)}$[19] given as:

$$E_{disp}^{D3} = E^{(2)} + E^{(3)} \tag{5}$$

The two-body term $E^{(2)}$ is expressed as:

$$E^{(2)} = -\sum_{n=6,8}\sum_{A<B} s_n \frac{C_n^{AB}(\mathrm{CN}^A, \mathrm{CN}^B)}{R_{AB}^n} f_{\mathrm{damp,n}}(R_{AB})$$

$$\mathrm{CN}^A = \sum_{B \neq A}^{N} \frac{1}{1 + \exp\left[-k_1\left(\frac{k_2(R_{A,\mathrm{cov}} + R_{B,\mathrm{cov}})}{r_{AB}} - 1\right)\right]}$$

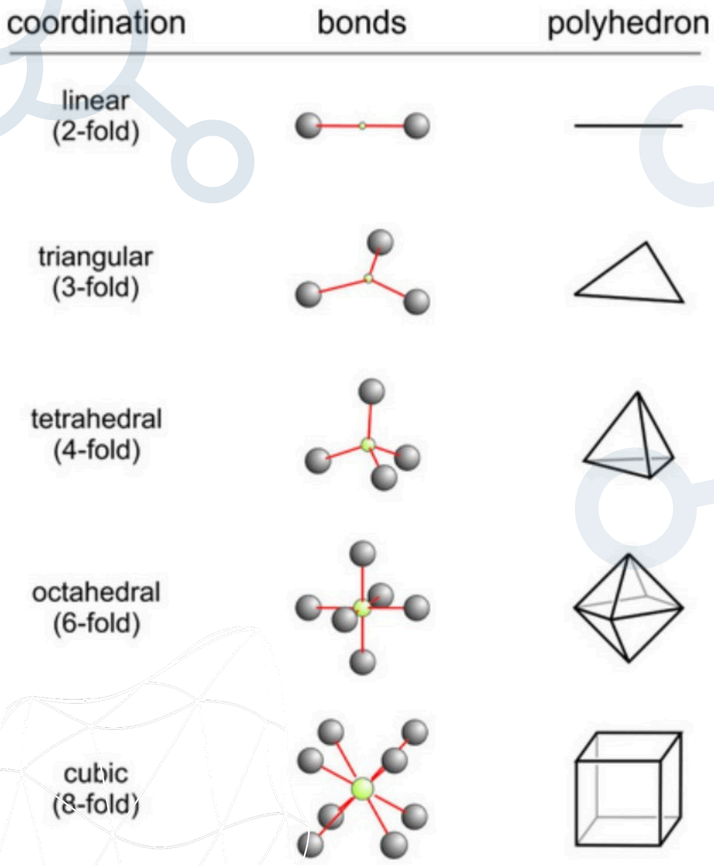
where N is the total number of atoms, $R_{A,\mathrm{cov}}$ and $R_{B,\mathrm{cov}}$ are scaled covalent radii of atoms A and B, and the constants $k_1 = 16$ and $k_2 = \frac{4}{3}$ are predefined.

$$C_6^{AB}(\mathrm{CN}^A, \mathrm{CN}^B) = \frac{\sum_i^A \sum_j^B C_{\mathrm{ref}}^{AB}(\mathrm{CN}_i^A, \mathrm{CN}_j^B) L_{ij}}{\sum_i^A \sum_j^B L_{ij}}$$

where C_{ref}^{AB} denotes precomputed reference dispersion coefficients for coordination environments CN_i^A and CN_j^B, and the weighting factors L_{ij} are defined by:

$$L_{ij} = \exp\left[-k_3\left((\mathrm{CN}^A - \mathrm{CN}_i^A)^2 + (\mathrm{CN}^B - \mathrm{CN}_j^B)^2\right)\right]$$

Coordination Number (CN) in Solids



Higher coordination number indicates denser atomic packing

- **Definition**
  - Number of nearest neighboring atoms around a central atom
- **Key Idea**
  - Describes the local atomic environment
  - Higher CN → more densely packed structure
- **Role in D3 Model**
  - Used to make dispersion environment-dependent
  - Helps improve accuracy of calculations
- **Example**
  - Low CN → isolated atoms High CN → dense crystal structures

Zero Damping

A widely used form of the damping function is the zero-damping variant of the D3 correction, denoted as D3(0), which is given by:

$$f_{\mathrm{damp},n}^{\mathrm{D3(0)}}(r_{ij}) = \frac{1}{1 + 6\left(\frac{r_{ij}}{s_{R,n} R_{0ij}}\right)^{-\alpha_n}}$$

where the parameters $\alpha_6 = 14$, $\alpha_8 = 16$, and $s_{R,8} = 1$ are fixed, while $s_{R,6}$ serves as a functional-dependent adjustable parameter.

DFT-D3(BJ) Model

- **Further Improvement** Introduces Becke -Johnson damping
- **Key Feature** Corrects short-range interaction errors
- **Advantage** Avoids overestimation at small distances Gives most accurate results

Most refined version of D3

An alternative and more widely adopted damping scheme is the Becke–Johnson damping function, used in the D3(BJ) variant:[22]

$$f_{\mathrm{damp},n}^{\mathrm{D3(BJ)}}(R_{AB}) = \frac{R_{AB}^{n}}{R_{AB}^{n} + (a_1 R_{0AB} + a_2)^n} \tag{8}$$

where a_1 and a_2 are empirical damping parameters optimized against high-level reference interaction energies.

MBD(Many-Body Dispersion)

The Many-Body Dispersion (MBD) method addresses these limitations by representing atoms as quantum harmonic oscillators (QHOs) that interact via long-range dipole– dipole couplings.

$$H_{\mathrm{MBD}} = \sum_i \left(\frac{p_i^2}{2m} + \frac{1}{2} m\omega_i^2 x_i^2 \right) + \sum_{i<j} V_{\mathrm{dip}}(x_i, x_j)$$

DFT-D3 vs MBD - Accuracy vs Practicality

DFT-D3

- Semi-empirical dispersion correction
- Fast and computationally efficient
- Widely used in solid-state studies
- Slightly lower accuracy

MBD

- Accounts for many-body interactions
- Higher accuracy in complex systems
- More physically rigorous
- Computationally expensive

MBD is more accurate, but DFT-D3 is preferred for efficiency and practicality

Objective of the Study

DFT-D3 Method → Apply to Materials → Compare with Data → Evaluate Accuracy

Data-driven evaluation of D3 accuracy in solid-state systems

Main Goal To evaluate the accuracy of DFT-D3 methods For solid-state materials

Key Focus

- Analyze how well D3 performs in:
  - Different structures
  - Different elements
  - Different environments

Approach

- Use a data-driven analysis
- Compare with reference data

Methodology

Computational Approach

Used Density Functional Theory (DFT)
Applied D3 dispersion correction

Data Set

Large set of solid-state materials
Includes different:

Elements(K, Cu, Li, Mg, Zn, Mn)
Crystal structures

Parameters Analyzed

Binding energies
Structural properties
Coordination numbers (CN)

Analysis Method

Compared results with reference data(MBD)
Performed statistical analysis error

Materials Dataset → DFT Calculation → Apply D3 → Compare with Reference → Error Analysis

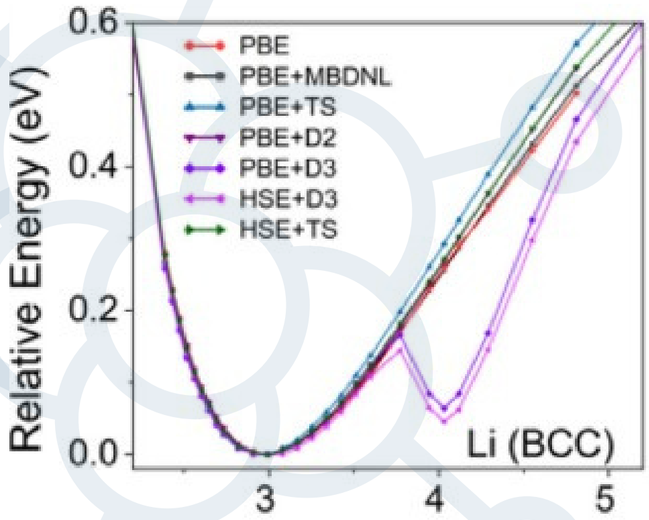
Workflow of computational and data-driven analysis

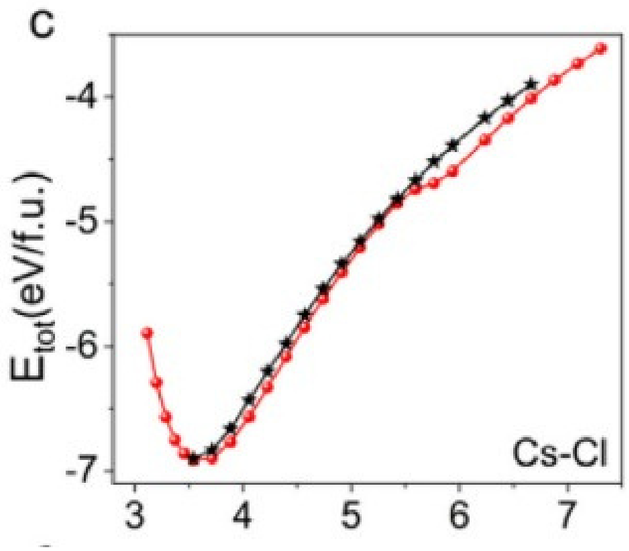
Energy vs Distance Analysis

Observation Energy varies with interatomic distance (R) Shows bond formation behavior

Key Finding D3 sometimes predicts multiple minima Indicates possible inaccuracies

Interpretation Correct curve → single stable minimum D3 curve → extra minima





Energy vs distance showing two minima behavior in D3

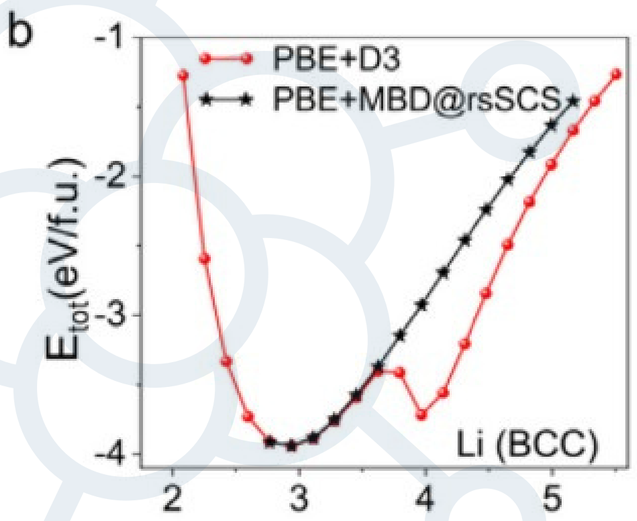
Conclusion D3 may introduce artifacts in energy prediction

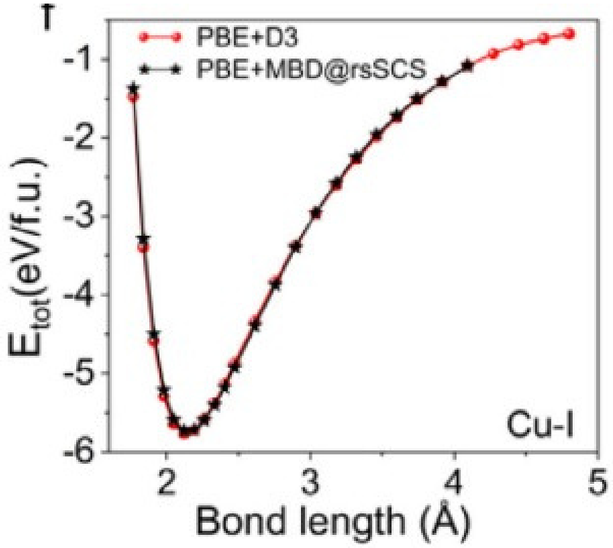
Case Study – Material-Specific Behavior

Observation
Different materials show different energy profiles Behavior depends on atomic structure

Example Some systems show abnormal multiple minima Especially in certain crystal structures (e.g., BCC)

Key Insight D3 performance is not uniform Strongly affected by local environment

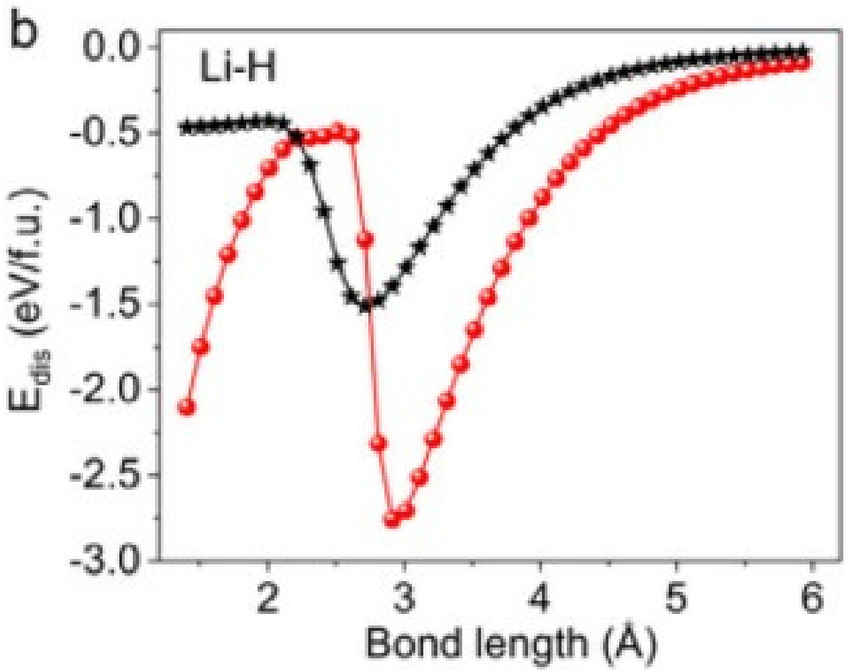




Variation in energy curves across different materials

Conclusion Material-specific analysis is essential for accuracy

Dispersion Energy vs Bond length



For systems like Li-H, they observed that the dispersion term itself shows a minimum. This confirms that the artifact originates from the dispersion correction not from the underlying DFT functional.

Origin of the artifact:

- **Unlike earlier dispersion models, D3 does not assign fixed C6 values to atoms and the dispersion coefficient**
- **depends on the coordination number(CN) of atoms A and B.This means the dispersion strength changes depending on the local chemical environment.**

Atoms in different environments have different:

- polarizabilities
- electronic screening
- bonding characteristics

Variation of C_6 with Coordination Number

What is C_6?

- Dispersion coefficient representinginteraction strength Determines magnitude of van der Waals
- forces

Observation

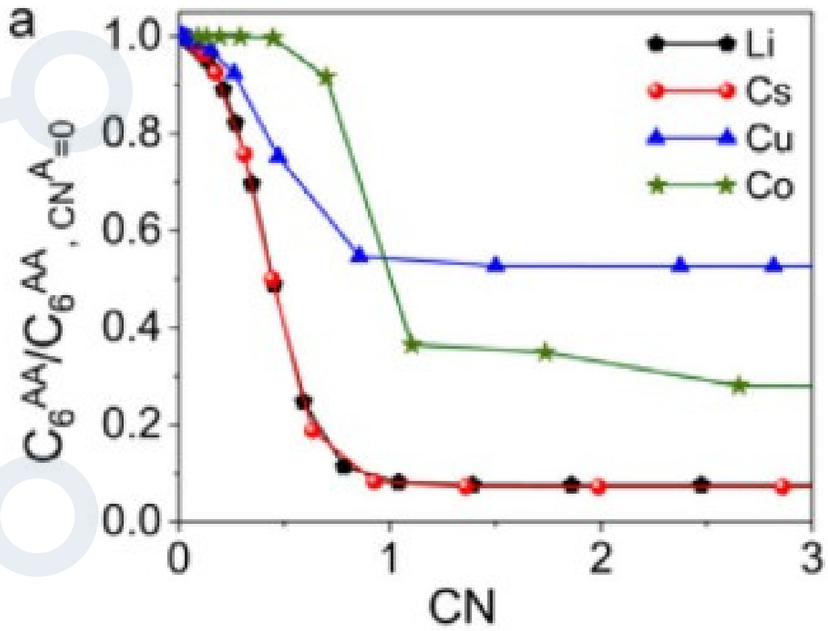
- C_6 varies with coordination number (CN)
- Not constant across environments

Key Insight

- Higher CN → modified C_6 values Reflects
- environment-dependent behavior

Impor tance

- Explains why D3 adapts better than D2 Improves accuracy in solid-state systems



Dependence of dispersion coefficient (C_6) on coordination number

Environment dependence is key strength of D3 model

Element-Specific Variation in C_6

Observation

- C_6 variation is different for different elements Not
- a universal trend

Key Insight

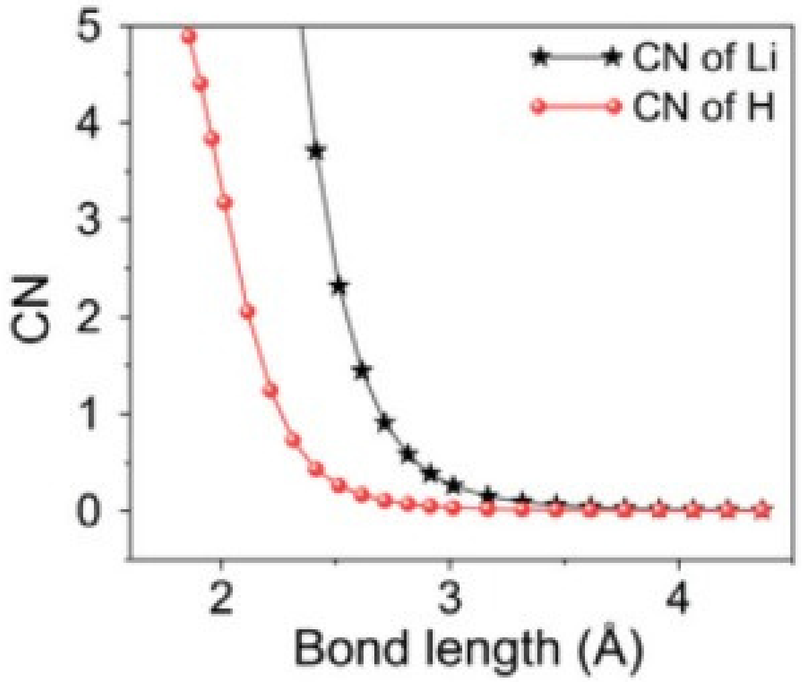
- Each element shows unique dependence on CN
- Indicates strong material-specific behavior

Interpretation

- Electronic structure affects dispersion strength
- Leads to variation in interaction energies

Conclusion

- D3 captures element-wise differences effectively

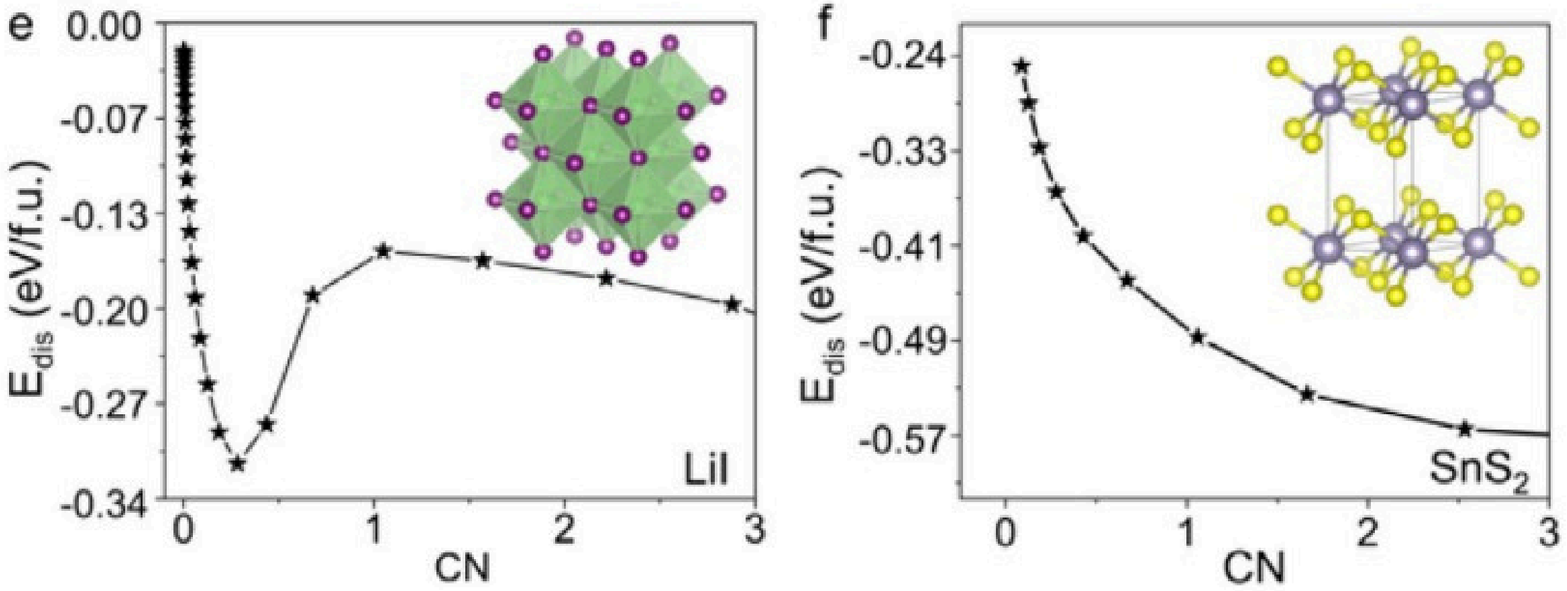


Element-dependent variation of C_6 with coordination number

Dispersion interactions are both environment- and element-dependent

Why Metals Are Most Affected and others are not?

Theartifactisstrongestforhighlypolarizableatoms,suchas:Lithium, Cesium



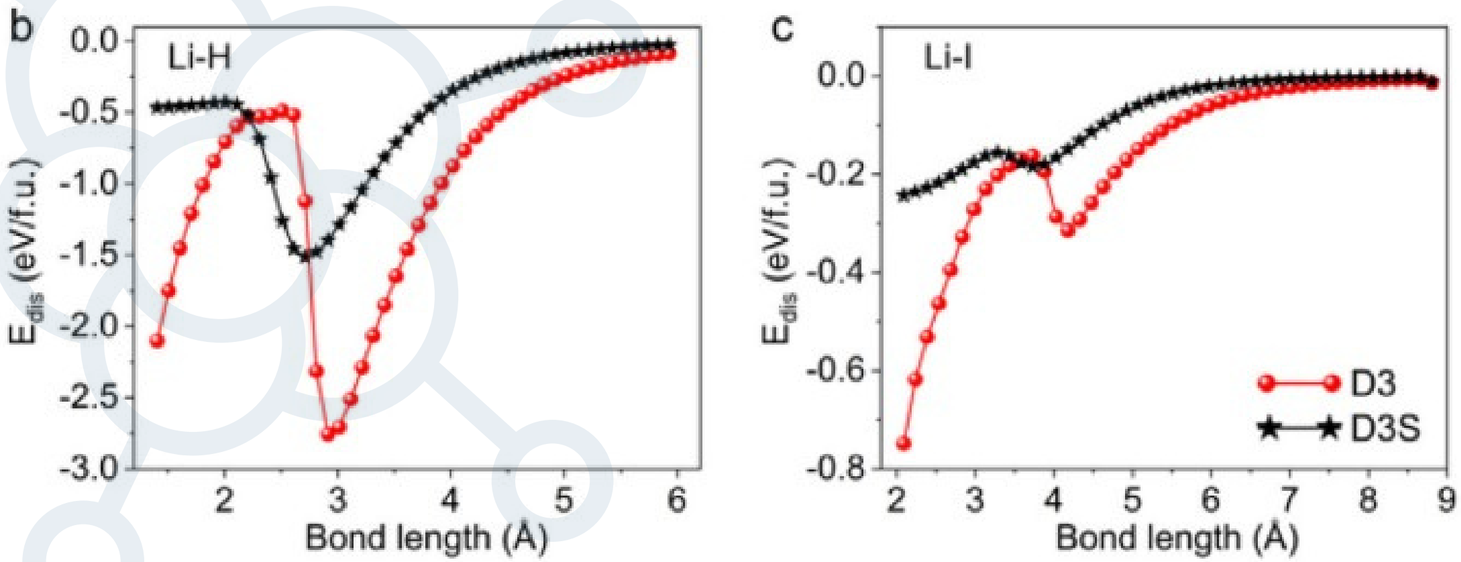
Highly polarizable atoms have large dispersion coefficients.
Therefore any abrupt change in C6 produces large changes in energy.
This is why metallic systems show the strongest artifacts.

Comparison of D3 and Improved Met hod(D3S)

D3S improves accuracy by correcting short-range dispersion errors present in D3

Key Improvement: Reduces multiple minima issue Better energy prediction

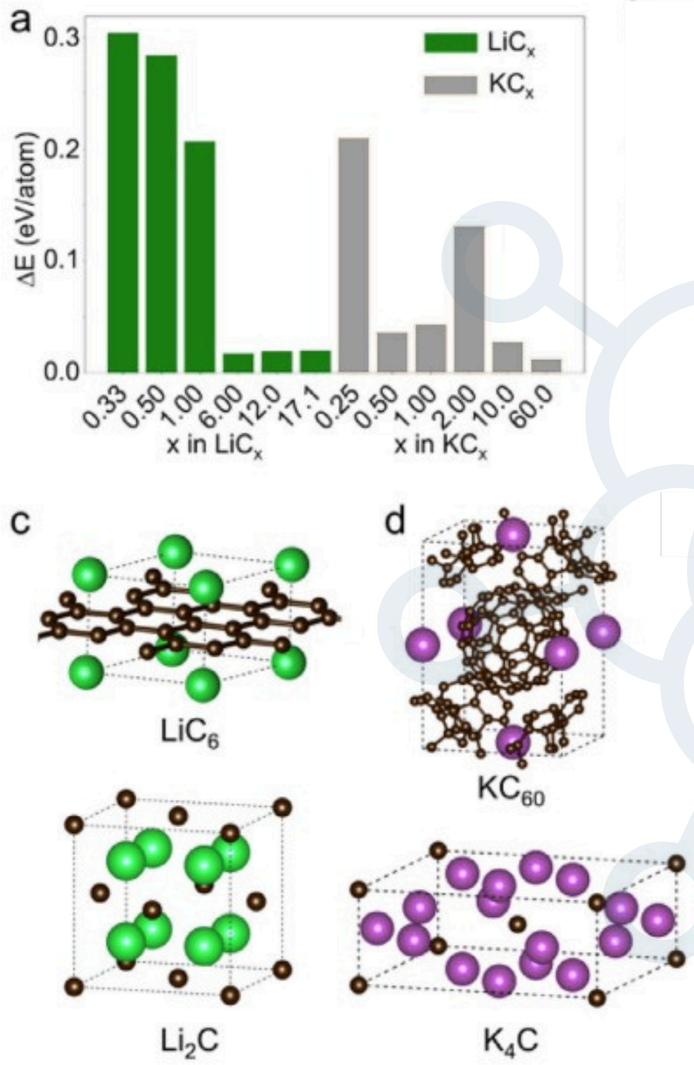
Interpretation: Improved model corrects D3 limitations



Improved method(D3S) reduce errors present in D3

Conclusion Enhanced methods provide higher reliability

Structural Impact on D3 Accuracy



Observation

- Different crystal structures show different D3 accuracy

Key Insight

- Dense structures → better prediction
- Open structures → larger errors
- D3 errors increase with metal content and decrease when the carbon content increases.

Interpretation

- Atomicpacking affects dispersion interactions
- stronger covalent bonding networks reduce artificial dispersion effects.

Key Findings of the Study

D3 Accuracy is System-Dependent

1. Error Behavior D3 shows large errors near equilibrium

2. Distance Dependence Accuracy varies with bond length

3. Structural Influence Dense structures → better accuracy Open structures → higher error

4. Model Improvement D3S significantly improves per formance

Conclusion

Understanding D3 D3 provides dispersion correction but has limitations

Observed Issues Errors near equilibrium Dependence on structure and distance

Improvement Achieved D3S reduces errors and improves accuracy

Systems where D3 works well:

covalent frameworks layered materials low metal-content structures.

Systems where D3 may fail:

metallic solids ionic compounds dense structures with highly polarizable atoms.

Thank You For Joining

We hope this presentation inspired curiosity, built understanding, and encouraged you to explore the world of computational science further.

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